Sambhav Jain | Software Engineer

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Education

Indian Institute of Technology, Madras Chennai, (TN) IN

Master of Technology (M.Tech) *Computer Science and Engineering* **Aug 2023 – May 2025**

CGPA: 8.68

Dayananda Sagar Academy of Technology and Management Bengaluru, (KA) IN

Bachelor in Engineering (B.E.) *Information Science and Engineering* **Jul 2017 – Aug 2021**

CGPA: 7.76

Work Experience

TVS Motors Bengaluru, (KA) IN

Software Engineer - VIC Team **Jun 2025 – Present**

Core ECU Software Development

- Developed embedded application software for automotive ECU using **C on S32K platform (S32DS IDE).**
- **Conducted functional and transient testing, delivering stable production binaries**
- **Collaborated with hardware and system teams for requirement alignment and issue resolution**

High-Side Driver (HSS) Control & Diagnostics

- Developed **high-side switch (HSS) control logic** for vehicle lighting system (TSL/hazard functionality)
- Implemented **fault detection and diagnostics** (open-load, latch status via Global Status Byte) with **real-time error reporting over CAN**
- Debugged and validated HSS behavior across operating conditions, ensuring **synchronization between hardware driver status and software control**

Low Power Mode & Vehicle State Management

- Designed and validated power state transitions (active, suspend, deep sleep), improving system efficiency and wake-up reliability.
- Implemented **wake-up handling and system re-initialization**, optimizing power consumption and ensuring stable state transitions.
- Developed **delayed shutdown logic (~15 sec)** and ensured safe de-initialization of peripherals (CAN, UART, timers) before sleep, reducing power consumption by **~20–30%**.
- Optimized **low-power transitions**, reducing system latency from **7s to 5s** and improving wake-up performance

Projects

RTOS-Based Automotive Peripheral Framework

- Designed low-level drivers for **UART and CAN**, using memory-mapped register access.
- Implemented **RTOS scheduling, ISR handling, and synchronization mechanisms (queues, semaphores)**
- Built modular architecture (HAL + abstraction layers) for scalable embedded systems
- Developed Python-based scripts for **basic test automation and validation workflows**.

Telemetry-based driving analytic dashboard (Python | Streamlit | Pandas)

- Developed a **real-time driving behavior analysis system** to classify and score rider performance using telemetry data.
- Built an end-to-end data pipeline, including **data cleaning, resampling, and feature engineering**, achieving **99.2% data stability**
- Designed adaptive event detection algorithms for **overspeeding, harsh braking, and throttle spikes**, achieving **>94% precision**.

Relevant Coursework & Certifications

- **TVS:** [Matersing RTOS], Mastering MicroController, [UML State Machine], [Embedded Driver Development], [Embedded C].
- **Mtech:** Network Security System, GPU Programming, Secure Systems Engineering.
- **Btech:** Computer Networks, Operating Systems, Programming for Problem Solving,

Skills

Embedded Systems: RTOS, Device Drivers, Communication Protocols (UART, I2C, CAN)